

A 60 year old male was admitted to the ICU 10 days previously with septic shock and acute lung injury following anastomotic dehiscence of an anterior colonic resection. He was recovering well on the HDU until 48 hours ago when his condition deteriorated with abdominal distension and diarrhoea. It is suspected that he has developed a new systemic inflammatory response syndrome.

A. How should the patient be assessed clinically? (60%)

B. List the key initial investigations in this patient? (40%)

(A)

This is a concerning situation as patient may be in septic shock (recent episode 10 days ago in ICU).

Possible causes include:

Abdominal: Anastomotic leak, bowel dehiscence, intestinal obstruction, infected intra-abdominal collections, surgical wound infection

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Respi: Hospital acquired pneumonia (HAP)

Other occult infections (line infections, UTI, pressure sores)

I will attend to this patient as soon as possible, focusing my assessment on:

- Causes for deterioration, abdominal distension, and diarrhoea
- Which may require urgent abdominal scans and possible surgical intervention.
- Evidence for new systemic inflammatory response syndrome (SIRS)
- Haemodynamic stability of patient for scans/ intervention
- Need to up-triage disposition to ICU.

Surgeons should also be informed to assess patient.

History

- Review patient's symptoms – fever, shivering, cough/ SOB, abdominal pain - nature/ onset/ duration/ pain score/ location/ radiation/ associated symptoms (distension, diarrhoea, constipation with spurious diarrhoea).
- Check for fever/ shivering
- Review surgical op notes (Any intra-op complications/ concerns during anastomoses? Need for diverting ileostomy/ colostomy?)
- Intake-output (IO) charts (Intake – oral? TPN?, reduced PU, BO – bristol stool chart, frequency)
- Any abnormal drain/ stoma output – volume, consistency, colour, presence of blood/ melaena?
- Review previous cultures and sensitivities, appropriateness of antibiotics.
- Review medications (laxatives, which may can diarrhoea/ C.Diff,

Physical Examination

- Airway (ability to speak in full coherent sentences, presence of stridor)
- Breathing (RR, laboured breathing, use of accessory muscles, auscultate for crepitation/ wheeze, decreased or unequal air entry)
- Circulation (Signs of dehydration/ hypoperfusion/ vasodilation – warm peripheries, feel pulse, capillary refill time (CRT), mucosal membranes, skin turgor)
- Evaluate GCS - any altered mental status, confusion or lethargy. Any localising neurological signs.
- Get full set of vitals (BP, HR, SpO2, ECG, temp)
- Examine abdomen – sites for tenderness/ guarding/ surgical site wound healing, abdominal drains/ stoma if present)
- Measure and trend intra-abdominal pressure (IAP)
- Examine existing lines (IV, IA, CVP, PICC, IDC) for signs of infection

(B)

Key initial investigations include:

FBC – Trend TW, Hb, Plt, leukocytosis

RP – Check and ensure electrolytes replete. Monitor for AKI.

CRP/ Procal – Uptrending inflammatory markers may suggest worsening bacterial infection.

LFT – Hepatic involvement/ cause for sepsis.

Blood/ fluid/ respi cultures – Should be taken prior to antibiotics escalation.

GXM – In prep for possible surgical intervention.

POCT ABG + lactate – oxygenation/ acid-base/ perfusion status

POCUS Heart/Lung – Evaluate cardiac function, fluid status, lung consolidation/ effusions/ interstitial oedema.

CXR – Evaluate for new HAP/ worsening ALI/ diaphragmatic splinting from intra-abdominal distension/ air under diaphragm

CTAP – Evaluate for intra-abdominal collections, anastomotic leak, bowel dehiscence, and intestinal obstruction.